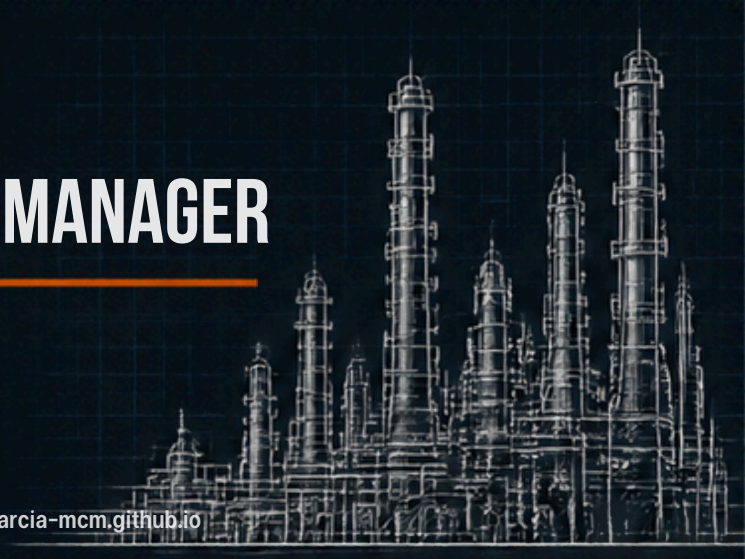
NÉSTOR GARCÍA —
MECHANICAL CONSTRUCTION MANAGER



International site execution leader across Oil & Gas, Petrochemical, Power Generation, Renewable and Biofuels performing roles from QA/QC Manager to Site Manager and Client Representative across Europe, Africa, the Middle East and North America.

Valencia, Spain • Available for new projects.

Contact: nestorgarcia.emp@gmail.com | +34 627 225 615 | Portfolio: nestorgarcia-mcm.github.io
linkedin.com/in/ngarcia-mechanical-construction-manager

Oil & Gas | Petrochemical | Power Generation | Renewables & Biofuels | Recycling | Mission-Critical / Hyperscale

34
total projects

23.3
yrs years of experience

2001–2026
career span

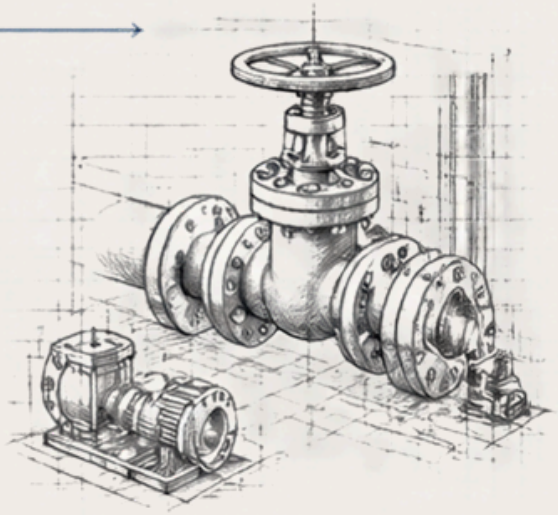
50%
of projects Greenfield

19
projects as final warranty stamp responsible

17
complex revamp & retrofit projects.

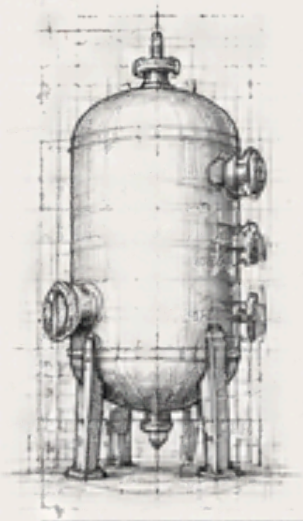
PROFILE

Mechanical Construction Manager with solid international experience in industrial and critical infrastructure projects – Oil & Gas, Petrochemical, Power Generation, Renewables & Recycling – performing key roles such as Mechanical Construction Manager, Client Representative, Senior Mechanical Supervisor, QA/QC Manager and Site Manager, leading Greenfield and Brownfield projects in Europe, Africa, the Middle East and North America. Proven leader in site execution, vendor surveillance, QA/QC, contractor management, supervision, installation and commissioning of mechanical systems (refrigeration, process, chemical separation, water treatment, firefighting, storage and others), guaranteeing quality, safety and compliance with deadlines in critical environments, with advanced capabilities in databases, Power BI, SQL, Excel, digital construction workflows and AI-assisted reporting. Accustomed to working under strict construction standards in critical projects. Demonstrated ability to lead multicultural teams, solve problems in the field, and ensure safe, efficient and on-time delivery in hyperscale projects.



RECENT EXPERIENCE

2025 - AUG 2026

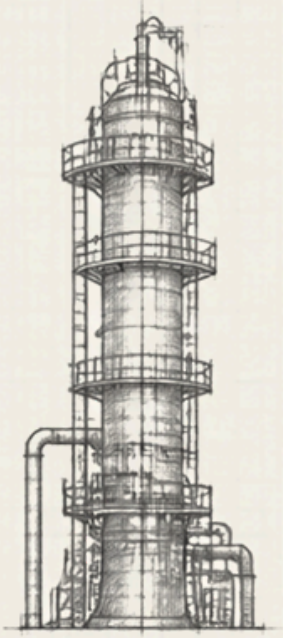


SENIOR MECHANICAL CONSTRUCTION MANAGER | ALDESA INGENIERÍA — PICASSO PROJECT, MOEVE •

La Rábida Energy Park, Palos de la Frontera (Huelva), Spain · Engineering: Worley.

- Construction of 2 units for Moeve at Southern Europe's largest second-generation (2G) biofuels complex, designed to produce 500,000 t/year of Sustainable Aviation Fuel (SAF) and renewable diesel (HVO) from organic waste.
- 500,000 t/yr SAF & HVO • 250K pipe-welding inches • 2 units • Power BI forecast over 35,017 material codes.
- Erection of static and dynamic equipment: flexible-configuration reactors, heat exchangers, fractionation columns and high-pressure pumping systems.
- Mechanical integration of flexible reactor units, designed to switch flow routing and operating conditions between HVO and SAF production.
- Integration of pre-assembly package units, organic waste pretreatment systems and a storage tank farm connected to marine logistics infrastructure.
- Project with high technological complexity, automation and modular construction.
- Prefabrication and installation of 250K pipe-welding inches, structural and static/rotating equipment, insulation, and Electrical & Instrumentation (E&I).

RECENT EXPERIENCE



2023 – 2025 | PROJECT MANAGER / SENIOR MECHANICAL TECHNICIAN | SULZER CHEMTECH, TOWER FIELD SERVICES — EUROPE • ÁFRICA • MIDDLE EAST.

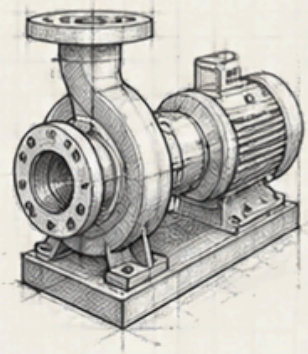
Tower Field Services representative and specialist technician, certifying works and advising the client through to warranty sign-off across multiple projects under the Total Tower Care model.

- **Services delivered:** Mass transfer internals — installation and replacement of conventional, high-performance and cartridge trays, structured and random packings, distributors, collectors and mist eliminators. **Revamps & retrofits —** mass transfer equipment replacement and column revamps, including mechanical management and QA/QC of a crude tower revamp, with milestones completed ahead of schedule. **Turnarounds & shutdowns —** pre-shutdown engineering, turnaround schedule planning and vessel integrity restoration. Emergency **response** — emergency replacement of column internals and emergency plant repairs on static equipment. **Automated weld overlay —** corrosion and erosion protection to extend vessel service life. **Engineering & technical support —** feasibility and debottlenecking studies to improve column performance.

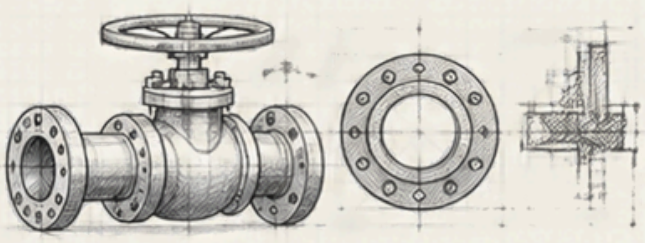
- **Responsibilities & results:** Certified completed works and advised the client from execution to final handover. Signed off 5-year warranties on completed weld overlay works and repairs. Managed simultaneous projects across Europe, Africa and the Middle East. Ensured full compliance with API and ASME standards, meeting all safety and quality KPIs. Increased client satisfaction by **50%** by delivering projects weeks ahead of schedule.

2022 – 2023 | MECHANICAL CONSTRUCTION MANAGER | TÉCNICAS REUNIDAS — ALBACETE, SPAIN.

Mechanical construction management of the Repetco Innovations PET/PE recycling plant, a highly complex industrial facility built around self-contained mechanical package units integrated into the overall process, with detailed design and mechanical interconnection services coordinated by Técnicas Reunidas. Identified and resolved construction issues on-site to prevent delays; monitored subcontractor operations and applied time, resource and cost-management strategies to deliver under-budget completion.



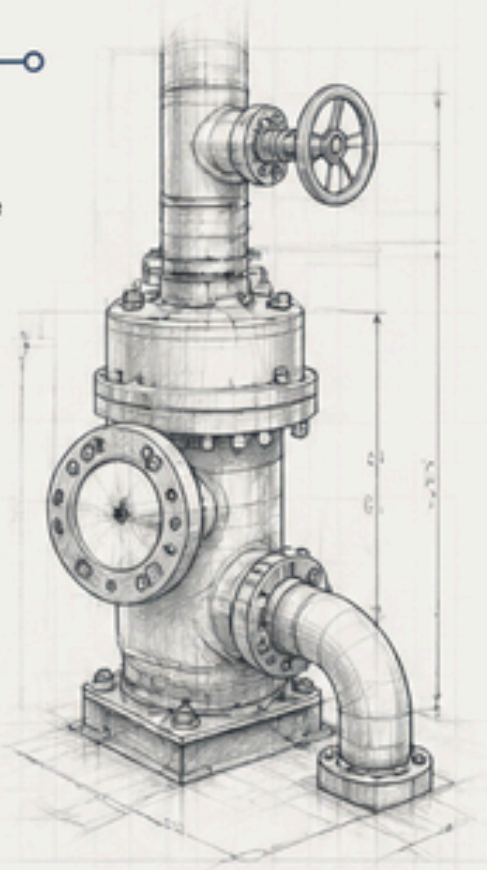
- **Mechanical packages: Main process packages (plastic treatment) —** crushing and milling units reducing post-consumer trays and containers to homogeneous flake size; patented mechanical delaminating modules separating the PE layer from PET with controlled friction, temperature and agitation, no polluting solvents; washing lines and densityseparation tanks or hydrocyclones.
- **Auxiliary services & heat exchange —** pressure vessels and heat exchangers managing water and thermal fluid flows; storage silos with dosing, discharge and pneumatic conveying of granulated material; industrial HVAC and ventilation for air quality and emissions control.
- **Protection & fluids —** fire protection systems (pumping networks and water storage around the plant perimeter, meeting plastic-waste safety regulations); process water treatment modules (filtration and mechanical purification) recirculating wash water and minimising water consumption.



NÉSTOR GARCÍA / MECHANICAL CONSTRUCTION MANAGER

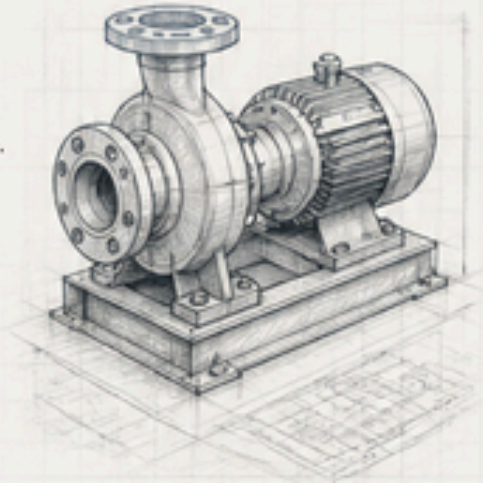


CORE RESPONSBILITIES



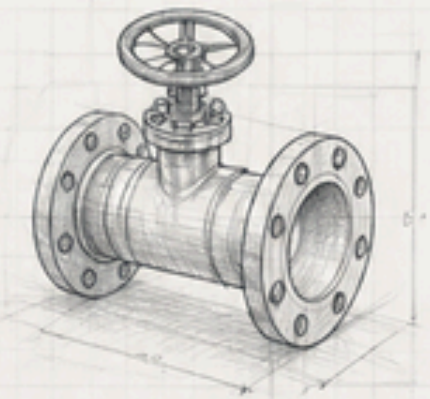
- **Site leadership & delivery** — lead multidisciplinary site teams, subcontractors and vendors to deliver mechanical scope on schedule and to quality standards.
- **Mechanical execution** — piping systems, HP/MP piping, static and rotating equipment, pressure vessels, mass transfer columns, boilers, heat recovery and refrigeration systems (chillers, CRAH/CRAC, pumps, exchangers, cooling towers, firefighting, CHW/CWS networks, water treatment and auxiliary equipment).
- **QA/QC** — weld inspection, support & alignment checks, pressure testing, flushing, chemical cleaning and material verification.
- **Daily coordination** — with contractors, engineering, HSE, QA/QC and commissioning to ensure correct execution of MEP systems.
- **Technical review** — of P&IDs, isometrics, assembly drawings, ITPs, WPS/PQR, installation procedures and test methods.
- **Management** — of NCRs, punch lists, RFIs, technical deviations and field interference resolution.
- **Commissioning support** — functional testing, start-ups, hydraulic balancing, ΔT validation and thermal performance.
- **Planning & progress control** — logistics and monitoring of mechanical milestones on site.
- **Vendor surveillance** — in international workshops for mechanical equipment, ensuring compliance with ASME, EN and customer specifications.
- **Technical representation** — in front of international clients, engineering teams and suppliers.

TECHNICAL & DIGITAL SKILLS



- **Construction & Engineering** — Full mechanical package: piping, rotating & static equipment, high-pressure components, skid systems, pressure vessels.
- **Cooling & Mission-Critical MEP** — Chilled water systems, condenser cooling, heat rejection systems, redundancy strategies (N+1 / N+2) and commissioning protocols for mission-critical uptime.
- **Digital Construction & BIM** — Practical integration of Revit and Navisworks into large databases: parameter extraction, ETL workflows, 4D/5D coordination and Power BI visualization to support prefabrication and clash mitigation.
- **Contract & Cost Control** — Cost baseline development, earned value concepts for mechanical scope, change-order workflows, commercial risk registers and mitigation plans, SQL-backed cost reconciliation and variance reporting.
- **Process & Tooling Innovation** — Digitalization of inspections, material & equipment tracking systems, integration of construction data into centralized platforms, and development of proprietary progress / productivity / risk-control systems connecting contractors, engineering, QA/QC and commissioning.

DATA & REPORTING



- **Power BI** — dynamic dashboards: MTO vs. progress, QA/QC heatmaps, punch-list status, cost vs. progress.
- **Oracle & SQL** — database modelling, queries and reporting for MTO and asset traceability.
- **MS Access** — process modelling and automation, integrated with construction site data.
- **Excel (advanced)** — Power Query, VBA and pivot modelling for reconciliations and automation.

MATERIALS & CODES

Standards: ASME I / II / V / VIII / IX • ASME B31.1 / B31.3 / B31.4 / B31.8 • API • UNE-EN • EN • ISO 9001 / 14001 • DIN EN • ASHRAE • EN 378 • RITE •

Materials: Carbon Steel | Stainless Steel | Alloy Steels | Inconel | P22 | P91

EDUCATION

B.Sc. Computer Databases IT – Valencia Polytechnic University, Spain – 2001.

Graduate Certificate of Education (GCE) – I.B. Ramon Llull, Valencia, Spain – 1997.

CERTIFICATIONS

Inspector of Welded Construction, Level II (07/2016).

CFR – Certified First Responder (03/2020).

Mass Transfer Technologies (08/2023).

LANGUAGES

Spanish — Bilingual / Proficient (C2)	5/5
English — Advanced (C1)	4/5
French – Intermediate (B1)	3/5
Portuguese – Intermediate (B1)	3/5
Italian – Elementary (A2)	1/5

TRACK RECORD — YEARS BY INDUSTRY SUB-SECTOR

Petrol Refining	6.2 yrs · 14 proj.
Bio Refining	4.3 yrs · 5 proj.
Energy (power plants)	4.0 yrs · 5 proj.
Waste to Energy	3.75 yrs · 3 proj.
Solar	2.5 yrs · 2 proj.
Petrochemical	1.55 yrs · 4 proj.
Metal Production	1.0 yrs · 1 proj.

PROJECTS BY REGION · 10 COUNTRIES

Europe 29 · Africa 2 · Middle East 2 · North America 1

EXTENDED CODES & STANDARDS

ASME B16.5 / B16.9 / B16.20 / B16.34 · B31.5 / B31.9 / B31.11 / B31.12 / B31Q · API 610 / 650 / 1104 / 12J · ISO 3183 / 15156 · UNE-EN 10208 / 10255 / 13480 · UNE-EN-ISO 3183 / 15156 / 21809 · DIN EN ISO 3183 / 15156 / 19008 · DIN EN 10208 / 13480

DATA, DIGITAL & AI

Self-built SQL project database (34 projects, about 140,000 documents) connected to Power BI and published at nestorgarcia-mcm.github.io · AI assistants (Claude, ChatGPT, Microsoft 365 Copilot) for specification, datasheet and contract review, correspondence triage and reporting, with human sign-off · Excel Power Query / VBA and Python data scripts (AI-assisted).